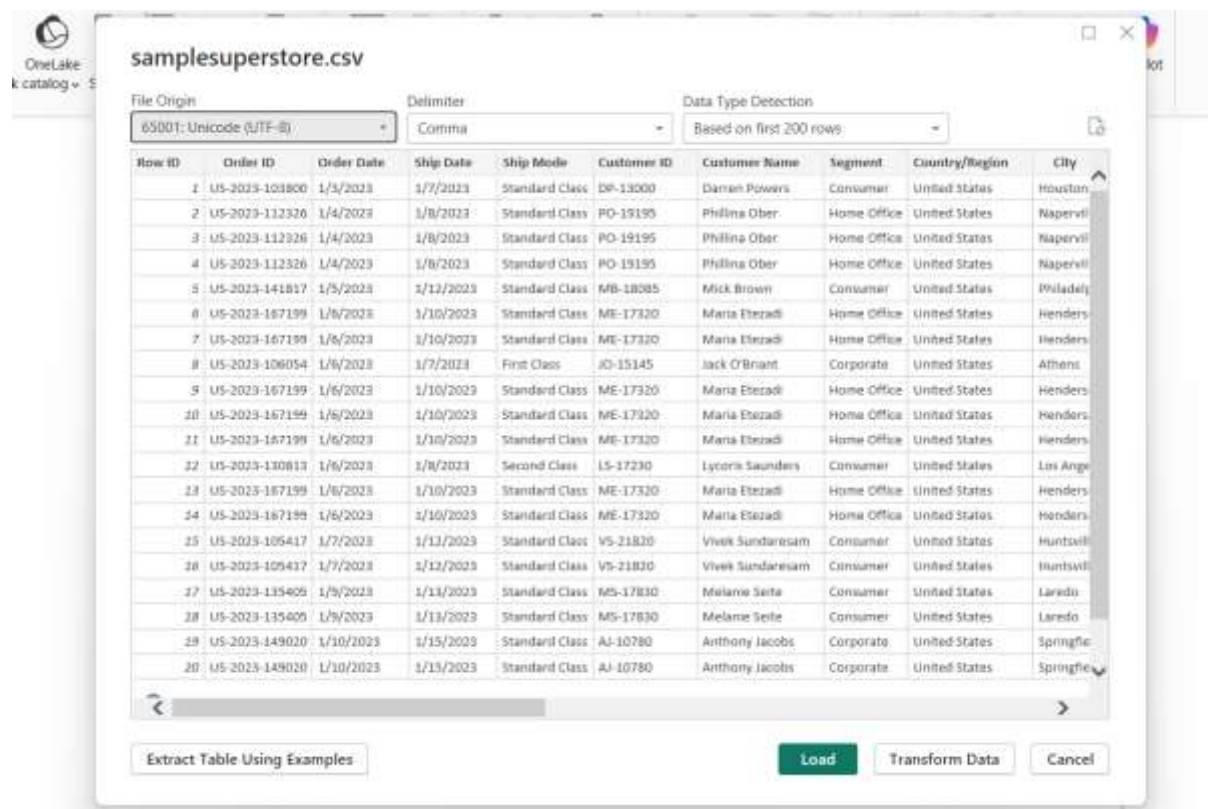


Week 1 Report: Superstore Sales Data Import, Power Query & Data Cleaning

Introduction

In this week's activity, I worked with the **Superstore Sales dataset** using Power BI Desktop. The main purpose was to learn how to **import raw data, clean it, and transform it** so that it becomes suitable for analysis. This step is called **data preprocessing**, and it is very important because dashboards and reports are only meaningful if the underlying data is accurate, consistent, and well-structured. The dataset represents sales transactions of a retail organization, including details about customers, products, regions, and profitability. By preparing this dataset properly, we set the foundation for deeper insights in the coming weeks.



Step 1: Importing Data into Power BI

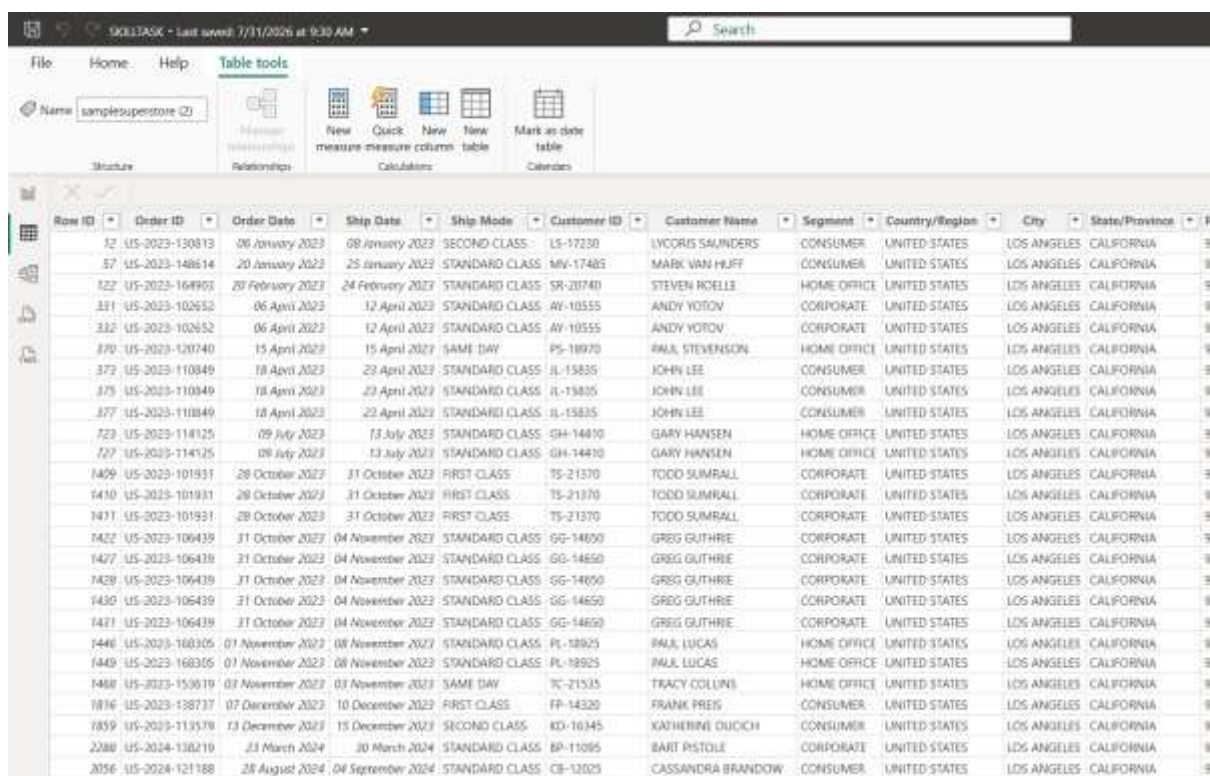
I opened **Power BI Desktop** and selected the option to import data from a CSV file.

The Superstore dataset was loaded successfully, and I could see the raw table in the **Report View, Data View, and Model View**.

These three views helped me understand the dataset from different angles:

- **Report View** shows how visuals will look.
- **Data View** displays the actual rows and columns.
- **Model View** shows relationships between tables.

This step gave me a first impression of the dataset and its structure.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City	State/Province	P
12	US-2023-130813	06 January 2023	08 January 2023	SECOND CLASS	LS-17230	LYCORIS SAUNDERS	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
57	US-2023-148614	20 January 2023	25 January 2023	STANDARD CLASS	MV-17485	MARK VAN HUFF	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
122	US-2023-164901	20 February 2023	24 February 2023	STANDARD CLASS	SR-20740	STEVEN ROELLE	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
311	US-2023-102652	06 April 2023	12 April 2023	STANDARD CLASS	AV-10555	ANDY YOTOV	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
332	US-2023-102652	06 April 2023	12 April 2023	STANDARD CLASS	AV-10555	ANDY YOTOV	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
370	US-2023-120740	15 April 2023	15 April 2023	SAME DAY	PS-18970	PAUL STEVENSON	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
372	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	IL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
375	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	IL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
377	US-2023-110849	18 April 2023	23 April 2023	STANDARD CLASS	IL-15835	JOHN LEE	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
723	US-2023-114125	09 July 2023	13 July 2023	STANDARD CLASS	GH-14410	GARY HANSEN	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
727	US-2023-114125	09 July 2023	13 July 2023	STANDARD CLASS	GH-14410	GARY HANSEN	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1409	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1410	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1411	US-2023-101931	28 October 2023	31 October 2023	FIRST CLASS	TS-21370	TODD SUMRALL	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1422	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1477	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1428	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1430	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1471	US-2023-106439	31 October 2023	04 November 2023	STANDARD CLASS	GG-14650	GREG GUTHRIE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1440	US-2023-160305	01 November 2023	08 November 2023	STANDARD CLASS	PL-18925	PAUL LUCAS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1449	US-2023-160305	01 November 2023	08 November 2023	STANDARD CLASS	PL-18925	PAUL LUCAS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1468	US-2023-153019	08 November 2023	08 November 2023	SAME DAY	TC-21535	TRACY COLLINS	HOME OFFICE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1816	US-2023-138737	07 December 2023	10 December 2023	FIRST CLASS	FP-14320	FRANK PREIS	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
1859	US-2023-113579	13 December 2023	15 December 2023	SECOND CLASS	KD-10345	KATHERINE DUICHI	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9
2288	US-2024-138219	23 March 2024	30 March 2024	STANDARD CLASS	BP-11095	BARIT PISTOLE	CORPORATE	UNITED STATES	LOS ANGELES	CALIFORNIA	9
2056	US-2024-121188	28 August 2024	04 September 2024	STANDARD CLASS	CB-12023	CASSANDRA BRANDOW	CONSUMER	UNITED STATES	LOS ANGELES	CALIFORNIA	9

Step 2: Inspecting Data in Power Query Editor

- I opened the **Power Query Editor** to explore the dataset in detail.
- I checked each column's **data type** (text, number, date, etc.) to ensure they were correct.
- I looked for **missing values** or blank records that could affect analysis.
- I verified whether records were **unique** or if duplicates existed.

- This inspection helped me identify what kind of cleaning and transformations were needed.

The screenshot shows a data transformation tool interface. The main table has columns: Row ID, Order ID, Order Date, Ship Date, and Customer ID. A context menu is open over the 'Ship Date' column, showing options like 'Copy', 'Remove', 'Duplicate Column', 'Change Type', etc. The 'Change Type' option is selected, and a sub-menu is open showing various data types like 'Decimal Number', 'Fixed decimal number', 'Whole Number', etc. The 'Date' option is selected. On the right, the 'APPLIED STEPS' panel shows a list of steps: 'Source', 'Promoted Headers', 'Changed Type', 'Changed Type with Locale', and 'Changed Type with Locale'.

Row ID	Order ID	Order Date	Ship Date	Customer ID
1	1/US-2023-108000	01-01-2023		
2	2/US-2023-112538	04-01-2023		
3	3/US-2023-112326	04-01-2023		
4	4/US-2023-112538	04-01-2023		
5	5/US-2023-104817	01-01-2023		
6	6/US-2023-107198	06-01-2023		
7	7/US-2023-107198	06-01-2023		
8	8/US-2023-108004	06-01-2023		
9	9/US-2023-107198	06-01-2023		
10	10/US-2023-107198	06-01-2023		
11	11/US-2023-107198	06-01-2023		
12	12/US-2023-108013	06-01-2023		
13	13/US-2023-107198	06-01-2023		
14	14/US-2023-107198	06-01-2023		
15	15/US-2023-105417	07-01-2023		
16	16/US-2023-105417	07-01-2023		
17	17/US-2023-105405	09-01-2023		
18	18/US-2023-105405	09-01-2023		
19	19/US-2023-140010	10-01-2023		
20	20/US-2023-140010	10-01-2023		
21	21/US-2023-105093	11-01-2023		
22	22/US-2023-107347	13-01-2023		
23	23/US-2023-108012	13-01-2023		
24	24/US-2023-107347	13-01-2023		
25	25/US-2023-118093	13-01-2023		

Step 3: Column Transformations

- I performed several transformations to make the dataset cleaner and easier to use:
 - Renamed columns to make them more understandable.
 - Split text fields where necessary (for example, separating combined information).
 - Trimmed spaces from text values to avoid mismatches.
 - Changed data types so that numbers, dates, and text were properly recognized.
- These transformations improved the dataset's consistency and readability.

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Query Settings

Layout Data Preview Columns Parameters Advanced Dependencies

Queries [1]

samplesuperstore (2)

Row ID Order ID Order Date

Valid 100% Valid 100% Valid 100%

Error 0% Error 0% Error 0%

Empty 0% Empty 0% Empty 0%

1	1	US-2023-103800	03-01-2023
2	2	US-2023-112326	04-01-2023
3	3	US-2023-112326	04-01-2023
4	4	US-2023-112326	04-01-2023
5	5	US-2023-141817	05-01-2023
6	6	US-2023-167188	06-01-2023

Step 4: Standardizing Categorical Attributes

- I focused on categorical fields such as Category, Sub-Category, and Segment.
- I ensured that text formatting was consistent (for example, all values in proper case, no extra spaces).
- This step is important because inconsistent text entries can cause errors in grouping and filtering during analysis.
- After standardization, categories became uniform and reliable for reporting.

AB	C Product Name	12	Sales	123	Quantity
%	Valid		Valid	100%	Valid
%	Error		Error	0%	Error
%	Empty		Empty	0%	Empty
	Message Book, Wirebound, Fo			16.448	
	GBC Standard Plastic Binding S			3.54	
	Avery 508			11.784	
	SAFCO Boltless Steel Shelving			272.736	
	Avery Hi-Liter EverBold Pen St			19.536	
	Global Deluxe High-Back Mana			3573.82	
	Rogers Handheld Barrel Pencil				
	Dixon Prang Watercolor Pencil				
	Ibico Hi-Tech Manual Binding S				
	Alliance Super-Size Bands, Ass				
	Southworth 25% Cotton Grani				
	Xerox 225				
	Wireless Extenders zBoost YX5				
	GE 30524EE4			391.98	
	Howard Miller 14-1/2" Diamet			76.728	
	Acco Four Pocket Poly Ring Bir			10.43	
	Newell 312			9.344	
	Memorex Micro Travel Drive 8			31.2	
	Howard Miller 11-1/2" Diamet			51.94	
	Avery 482			2.89	
	DAX Value U-Channel Document Frames, Easel Back			9.94	
	O'Sullivan Elevations Bookcase, Cherry Finish			333.999	
	Global Highback Leather Tilter in Burgundy			545.94	
	4009 Highlighters by Sanford			19.9	
	Avery Metallic Poly Binding			3.438	

Step 5: Creating Date Columns

- From the **Order Date** column, I extracted several useful time-based attributes:
 - **Year** – to analyze sales trends annually.
 - **Quarter** – to study seasonal performance.
 - **Month** – to compare monthly sales.
 - **Day** – to check daily transactions.
 - **Day of Week** – to see which days had higher sales.
- I used both the Power Query UI and M code to generate these new columns.

- These temporal attributes will be very helpful in time-series analysis and trend identification.

fx - Table.TransformColumns("#Changed Type with Locale1",{"Product Name", Text.Upper, type text}, {"Category", Text.Upper, type text})						
	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
100%	Valid	100%	Valid	100%	Valid	100%
0%	Error	0%	Error	0%	Error	0%
0%	Empty	0%	Empty	0%	Empty	0%
7-01-2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	HOUSTON
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL
2-01-2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	PHILADEL
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
7-01-2023	First Class	JO-15145	Jack O'Briant	Corporate	United States	ATHENS
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
8-01-2023	Second Class	LS-17230	Lyconk Saunders	Consumer	United States	LOS ANGE
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
2-01-2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	HUNTSVIL
2-01-2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	HUNTSVIL
3-01-2023	Standard Class	MS-17830	Melanie Seife	Consumer	United States	LAREDO
3-01-2023	Standard Class	MS-17830	Melanie Seife	Consumer	United States	LAREDO
5-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	SPRINGFI
5-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	SPRINGFI
4-01-2023	First Class	SV-20365	Seth Vernon	Consumer	United States	DOVER
8-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States	SAN FRAN
6-01-2023	Second Class	ND-18370	Natalie DeCherney	Consumer	United States	MOUNT P
8-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States	SAN FRAN

fx - Table.TransformColumns("#Changed Type with Locale1",{"Product Name", Text.Upper, type text}, {"Category", Text.Upper, type text})						
	Ship Mode	Customer ID	Customer Name	Segment	Country/Region	City
100%	Valid	100%	Valid	100%	Valid	100%
0%	Error	0%	Error	0%	Error	0%
0%	Empty	0%	Empty	0%	Empty	0%
7-01-2023	Standard Class	DP-13000	Darren Powers	Consumer	United States	HOUSTON
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL
8-01-2023	Standard Class	PO-19195	Phillina Ober	Home Office	United States	NAPERVIL
2-01-2023	Standard Class	MB-18085	Mick Brown	Consumer	United States	PHILADEL
0-01-2023	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	HENDERSK
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2-01-2023	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	HUNTSVIL
3-01-2023	Standard Class	MS-17830	Melanie Seife	Consumer	United States	LAREDO
3-01-2023	Standard Class	MS-17830	Melanie Seife	Consumer	United States	LAREDO
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5-01-2023	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	SPRINGFI
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8-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States	SAN FRAN
6-01-2023	Second Class	ND-18370	Natalie DeCherney	Consumer	United States	MOUNT P
8-01-2023	Standard Class	BD-11605	Brian Dahlen	Consumer	United States	SAN FRAN

Step 6: Loading the Clean Data Model

- After completing all transformations, I clicked **Close & Apply** in Power Query.
- The cleaned dataset was loaded back into Power BI.
- I checked the **Data View** again to confirm that the new columns and standardized values were present.
- The dataset was now ready for building dashboards and performing deeper analysis in future weeks.

